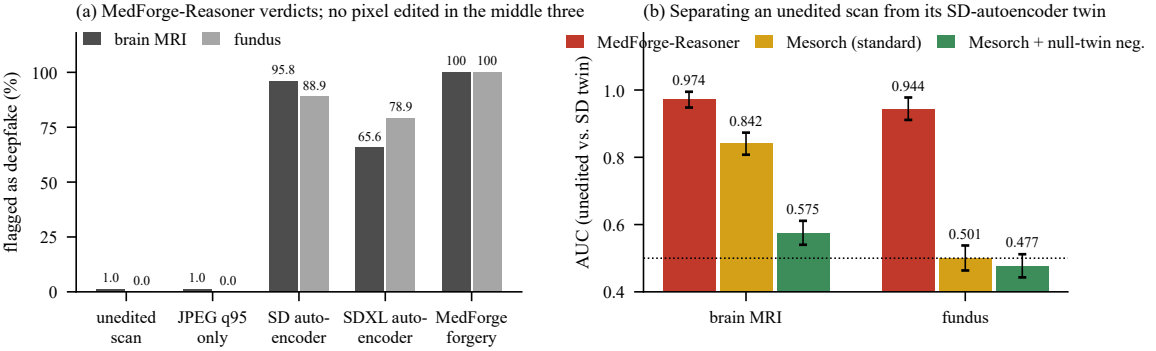
(a) MedForge-Reasoner verdicts; no pixel edited in the middle three
brain MRI
fundus
flagged as deepfake (%)
100
75
50
25
0
1.0
0.0
1.0
0.0
95.8
88.9
65.6
78.9
100
100
unedited scan
JPEG q95 only
SD auto-encoder
SDXL auto-encoder
MedForge forgery
(b) Separating an unedited scan from its SD-autoencoder twin
MedForge-Reasoner
Mesorch (standard)
Mesorch + null-twin neg.
AUC (unedited vs. SD twin)
1.0
0.8
0.6
0.4
0.974
0.842
0.575
0.944
0.501
0.477
brain MRI
fundus